

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1708

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

*I **Lokesh S (192511037)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- li. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- lii. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- Iv. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance}), \text{Vehicle}(\text{CarA}), \text{Behind}(\text{CarA}, \text{Ambulance})$

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) P1: $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$
- 2) P2: $\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$
- 3) P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
- 4) Facts: $\text{SoilDry} = \text{True}, \text{CropWheat} = \text{True}$

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).

(b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.

(c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) Stench [1,2] $\rightarrow$ Wumpus is adjacent to cell [1,2]
- 2) Breeze [1,1] $\rightarrow$ Pit is adjacent to cell [1,1]
- 3) Glitter [x,y] $\rightarrow$ Gold is at [x,y]
- 4) $\neg$ Wumpus [x,y] $\wedge$ $\neg$ Pit[x,y] $\rightarrow$ Safe[x,y]

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding ; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

CASE STUDY 1 — Smart Medical Diagnosis System

The knowledge base contains rules for diagnosing Flu, Measles and Pneumonia, along with facts about Patient A.

(a) Representation using Propositional Logic

Let the propositions be:

- **F** = Patient A has Fever
- **C** = Patient A has Cough
- **R** = Patient A has Rash
- **B** = Patient A has Breathlessness
- **Flu** = Patient A has Flu
- **M** = Patient A has Measles
- **P** = Patient A has Pneumonia

The rules can be represented as:

Rule 1:

Meaning: If the patient has Fever AND Cough, then the patient may have Flu.

Rule 2:

Meaning: If the patient has Fever AND Rash, then the patient may have Measles.

Rule 3:

Meaning: If the patient has Cough AND Breathlessness, then the patient may have Pneumonia.

Facts given:

Therefore, the initial knowledge base is:

(b) Forward Chaining

Forward chaining starts with the available facts and repeatedly applies rules whose conditions are satisfied.

Initial facts

Step 1 — Apply Rule 1

Rule 1:

We have:

Therefore:

Derived fact: Patient A has possible Flu.

Step 2 — Apply Rule 2

Rule 2:

We have:

But:

There is no fact stating that Patient A has Rash.

Therefore, **Rule 2 cannot fire.**

Step 3 — Apply Rule 3

Rule 3:

We have:

Therefore:

Derived fact: Patient A has possible Pneumonia.

Final Forward-Chaining Conclusions

Measles cannot be derived because Rash is not given.

(c) Backward Chaining for Pneumonia

Goal:

Find a rule that can conclude Pneumonia.

Rule 3 is:

Therefore, the goal is reduced into two subgoals:

Goal

Pneumonia(A)

|

Rule 3

/ \

Cough(A)

Breathlessness(A)

|

FACT

|

FACT

We know:

and

Therefore both subgoals are satisfied.

Hence:

Conclusion:

Patient A can be logically diagnosed with possible Pneumonia using backward chaining.

CASE STUDY 2 — Autonomous Traffic Management Agent

The FOL knowledge base contains rules for emergency vehicles, clear paths, green signals and vehicle movement.

(a) Unification

Rule 1

Facts:

We need to match:

with:

Therefore:

Similarly:

matches:

with the same substitution.

MGU

After substitution, Rule 1 becomes:

Both conditions are facts.

Therefore:

is derived.

(b) Resolution Proof for Proceed(CarA)

We need to prove:

Step 1 — Convert Rule 1 to CNF

Original:

Using:

becomes:

Therefore:

Step 2 — Rule 2

Original:

This gives two clauses:

and

Step 3 — Rule 3

CNF:

Step 4 — Relevant facts

Step 5 — Derive ClearPath(Ambulance)

From:

and Rule 1:

Resolution gives:

Step 6 — Derive GreenSignal(Ambulance)

Rule 2:

Since:

we obtain:

Step 7 — Derive Proceed(CarA)

Rule 3:

Given:

Resolution eliminates the first three negative literals.

Therefore:

Resolution Chain

Vehicle(Ambulance)



EmergencyType(Ambulance)



ClearPath(Ambulance)



GreenSignal(Ambulance)



Vehicle(CarA) + Behind(CarA,Ambulance)



Proceed(CarA)

Hence:

(c) Handling Two Simultaneous Emergency Vehicles

The current knowledge base is **partially sufficient but not complete** for handling two simultaneous emergency vehicles.

Suppose we have:

Because Rule 1 uses a universally quantified variable:

it can be applied independently to both vehicles.

Therefore:

and

can be derived.

However, the KB does not contain rules for **priority between two emergency vehicles**, collision avoidance, intersection conflicts, or deciding which vehicle should receive the green signal first.

Additional rules could include:

and:

Also, facts describing the locations, routes and relative positions of both emergency vehicles would be required.

Conclusion:

The existing FOL rules support multiple vehicles individually, but additional priority, conflict-resolution and routing rules are needed for simultaneous emergencies.

CASE STUDY 3 — Agricultural Expert Reasoning System

The agricultural KB contains rules for irrigation, drip method and crop risk.

Let:

- = SoilDry
- = IrrigationNeeded
- = CropWheat
- = ApplyDripMethod
- = CropAtRisk

(a) Convert into CNF

P1

Given:

Using:

Therefore:

This is already in CNF.

P2

Given:

Convert implication:

Using De Morgan's law:

Therefore:

P3

Given:

Convert implication:

Using De Morgan's law:

Therefore:

Facts

Given:

and

CNF representation:

Complete CNF

(b) Resolution Refutation to Prove ApplyDripMethod

Goal:

For resolution refutation, negate the goal:

Add it to the KB.

Clause 1

Clause 2

Clause 3

Clause 4

Clause 5

Clause 6 — Negated Goal

Resolution Step 1

Resolve:

with:

The complementary literals are and .

Therefore:

Resolution Step 2

Resolve:

with:

We obtain:

So:

Resolution Step 3

Resolve:

with:

Therefore:

Resolution Step 4

We already have:

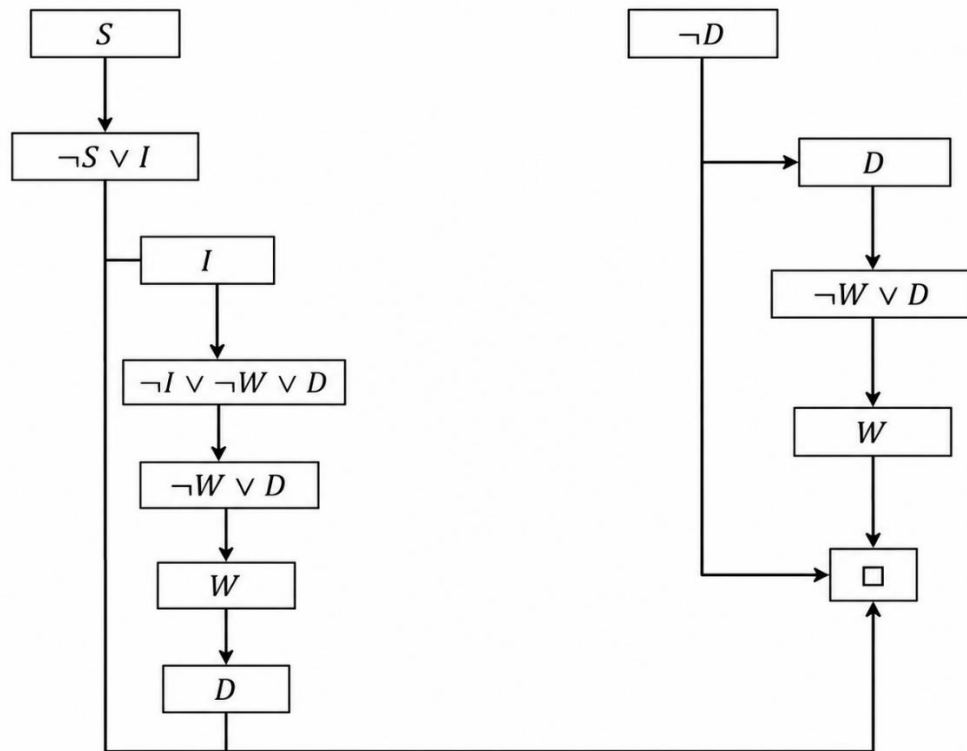
and the negated goal:

Resolve them:

Therefore:

The empty clause is obtained.

Resolution Tree



Since the empty clause is derived:

Therefore:

(c) Remove SoilDry

Initially:

was responsible for deriving:

through:

If SoilDry is removed, we no longer have:

Therefore Rule P1 cannot fire.

So:

cannot be derived.

Because P2 requires:

we also cannot derive:

Now consider P3:

We know:

But we **do not know**:

The absence of does **not** mean .

Therefore:

also cannot be derived.

Final result

Fact	Before removing SoilDry	After removing SoilDry
SoilDry	True	Unknown
IrrigationNeeded	True	Not derivable
ApplyDripMethod	True	Not derivable
CropWheat	True	True
CropAtRisk	Not derivable	Not derivable

Conclusion:

Removing SoilDry prevents the derivation of IrrigationNeeded and ApplyDripMethod. CropAtRisk still **does not follow**, because the KB does not provide .

CASE STUDY 4 — Wumpus World Knowledge Agent

The agent perceives:

- Stench at [1,2]
- Breeze at [1,1]
- Glitter at [2,2]

The provided rules relate these percepts to Wumpus, Pit and Gold locations.

(a) Belief State and Modus Ponens

Let:

- = Stench at [1,2]
- = Breeze at [1,1]
- = Glitter at [2,2]
- = Wumpus adjacent to [x,y]
- = Pit adjacent to [x,y]
- = Gold at [x,y]

Initial percepts

Step 1 — Stench

Rule:

Given:

Using Modus Ponens:

The adjacent cells include:

So the Wumpus is possible in one of these neighboring cells.

Step 2 — Breeze

Rule:

Given:

Therefore:

Possible adjacent cells are:

Step 3 — Glitter

Rule:

Given:

Therefore:

Derived Belief State

The agent knows:

The exact Wumpus and Pit positions cannot be determined from the given rules alone.

(b) Forward Chaining — Possible Wumpus and Pit Locations

Wumpus

Initial fact:

Apply Rule 1:

Therefore:

So the possible Wumpus locations are:

Pit

Initial fact:

Apply Rule 2:

Therefore:

So the possible Pit locations are:

Gold

Initial fact:

Apply Rule 3:

Therefore:

(c) Is the Path [1,1] → [1,2] → [2,2] Safe?

The proposed path is:

We must check each cell.

Cell [1,1]

From the Stench information:

Therefore [1,1] is a **possible Wumpus location**.

Also, the breeze at [1,1] tells us that a Pit is adjacent to [1,1], not necessarily inside [1,1].

Thus, the agent cannot establish that [1,1] is safe.

Cell [1,2]

From the Breeze:

Therefore [1,2] is a **possible Pit location**.

Hence [1,2] cannot be confirmed as safe.

Cell [2,2]

Glitter tells us:

However, the Stench rule indicates that [2,2] is also a possible Wumpus location:

Therefore [2,2] cannot be guaranteed safe.

Final Conclusion

The path:

is **not logically guaranteed to be safe**.

The reason is that:

- [1,1] may contain the Wumpus.
- [1,2] may contain a Pit.
- [2,2] contains the Gold but may also contain the Wumpus.

Therefore, the agent should **not assume the path is safe** based only on the available knowledge.